MLR780 Air Conditioning and Refrigeration 2021

Assignment 7
Duct System Design

Instructions

- This individual assignment is to be submitted online before midnight on **13 June 2021**.
- Repetitive calculation will be needed. Use a programming platform or a spread sheet of your choice. Use CoolProp for thermodynamic properties.
- Prepare a technical report that supplies all relevant equations, methods and input values, and results. Include everything that is to be graded in your report. Name it: "MLR780-2021-Assignment7-SURNAME.pdf"
- Each submission (PDF) **must** be accompanied by an official cover page obtainable from Click-UP, **as well as** your supporting documentation to prove you performed the analysis.
- The supporting documentation will not be graded, but will be checked: It must be a soft version of the script (a *.txt text file) or spread sheet with the following naming convention: "MLR780-2021-Assignment7-SURNAME.* " (The "*" represent the file extension, which will depend on the program you used – it should not be "pdf").
- **Only the PDF report will be graded.**
- Keep a time sheet documenting the time it took you to solve the questions and indicate this on the first page of your submission. Time spent on the lecture content / example / programming revision should not be included.

Introduction

Consider a duct system represented in Figure 1. Air is distributed to three identical diffusers (TROX HIVV, 198 mm inlet diameter) at zones (A), (B) and (C). Each diffuser has to supply 200 litres per second.

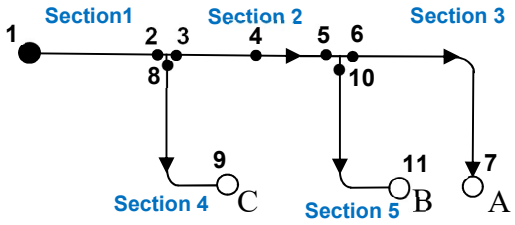


Figure 1 – Duct System Lay-out

The following ASHRAE fitting types (from ASHRAE 2009 Chapter 21) are used:

90° Bends:	CD3-5, Pleated.	(use subscript CD)
Diverging:	SD5-10 Tee.	(use subscript SD)
Diameter transition:	ED 4-1 Transition	(use subscript ED)
Enlargement:	SD4-1	(use subscript e)
Reducer:	SD4-2	(use subscript r)

The standard duct diameters (D) contained in Table 1 are available from a local duct manufacturer.

Table 1. Available duct diameters [mm]

100	225	400	700
125	250	450	800
150	275	500	900
175	300	550	1000
200	350	600	1200

Assume an absolute duct surface roughness of 0.09 mm, a uniform temperature of 14°C at 50% relative humidity through-out and an ambient pressure of 100 kPa. The relative available maximum total pressure (p_t) at point 1 is130 Pa (gauge). The straight duct length for each section is given in Table 2:

	L [m]
Section 1 (1,2)	8
Section 2 (2,3,4,5)	6
Section 3 (5,6,7)	11
Section 4 (2,8,9)	5
Section 5 (5,10,11)	5

Point 4 will be used in the last question, and until specified, is at some location along the straight duct that joins 3 and 5.

1.1 Obtain the static pressure drop (Δp_s) over the diffusers by using the catalogue available from click-UP.

1.2 Create a simulation model to calculate the relative pressure value (with point 1 as reference) of the total (p_t), velocity (p_v), and static pressure (p_s) values at each point in the duct system (point 1 to point 10). Make use of the indicated lengths and fitting types as defined by ASHRAE. Take note of the diffuser inlet diameter, and use the relevant component (from the list supplied previously) to connect the diffuser to the upstream duct portion. Use only **one duct** diameter per section (no SD4-1 or SD4-2 components for now). Supply a screenshot (if using a spreadsheet), or a programming script to prove that you created a model.

1.3 Use the **equal friction method** to specify the standard duct diameters. Start with an initial pressure gradient of 0.8 Pa/m and then iterate until you obtain a converged diameter set. For your final converged diameter set (final iteration), show the calculations for Section 4 in full (with equations). In addition, for each iteration (number your iterations), tabulate the following parameters for each section (1 to 5) in this sequence: $\frac{\Delta p}{L}, L, \dot{V}, V, D, \mathrm{Re}, f, K_{\mathrm{CD}}, K_{\mathrm{ED}}, C_{\mathrm{SD}}, L_{\mathrm{eq,CD}}, L_{\mathrm{eq,ED}}, L_{\mathrm{eq,SD}}, \Delta p_L, \Delta p_{\mathrm{CD}}, \Delta p_{\mathrm{ED}}, \Delta p_{\mathrm{SD}}, p_{\mathrm{in,t}}, p_{\mathrm{in,s}}, p_{\mathrm{out,t}}, p_{\mathrm{out,s}}$. Here K and C refer to the local pressure loss coefficients. All other symbols have their normal meanings, as was also used in the lecture example.

1.4 Plot the converged total and static pressure values (vertical axis) against the actual duct distance from point 1 (horizontal axis). Label all points on the diagram. Refer to the lecture example of the typical lay-out of such a diagram.

1.5 Update your design for Question 1.3 using the **static regain method**. Use a singular duct diameter for section 2 (i.e. ignore point 4 for now). For each subsequent iteration (starting from the converged result of question 1.3), tabulate the list of parameters (listed in question 1.3) for each section. No calculations or equations need to be shown.

1.6 Place point 4 a distance x from point 3 to divide section 2 into section 2a (the duct between 3 to 4) and section 2b (duct between 4 to 5). Introduce an enlargement / reducer at point 4, and select different diameters sections 2a and 2b (diameters D_{2a} and D_{2b} to better match the static pressure at point 5 with that at point 2. Consider different options of x (to the nearest 100 mm), D_{2a}, and D_{2b}. Use your previous selection of the duct diameter for section 1, and **only analyse section 2a and 2b** (Do not repeat the full duct design of question 1.3 or 1.5). Show relevant equations and tabulate your answers as follows:

x (m)	D_{2a}(m)	D_{2b} (m)	$\Delta p_s = p_{s,5} - p_{s,2}$(Pa)
…	…	…	…

Propose a selection for x, D_{2a}, and D_{2b} and motivate your selection by way of relevant plot to support your answer.

PLEASE INDICATE THE TIME YOU SPENT TO COMPLETE THIS ASSIGNMENT ON THE FIRST PAGE OF YOUR REPORT